

3.5 Homing Mode (HMM)

«Homing» describes the procedure according to which the drive seeks the home position (also called reference point or zero point). There are various methods to achieve this using limit switches at both ends of travel. Some of the methods use the index (zero) pulse train of an incremental encoder.



Annotation

Items marked with an asterisk (*) refer to internal values.

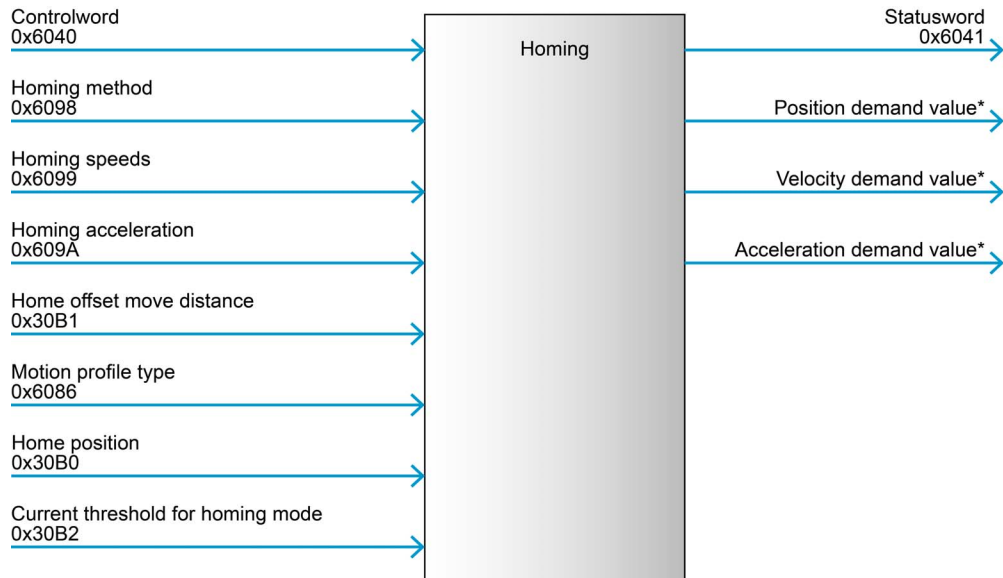


Figure 3-10 Homing Mode – Block diagram

3.5.1 Homing Trajectory Generator

The trajectory generator supports the following motion profiles. The movements are mode-dependent, the end positions will be calculated internally.

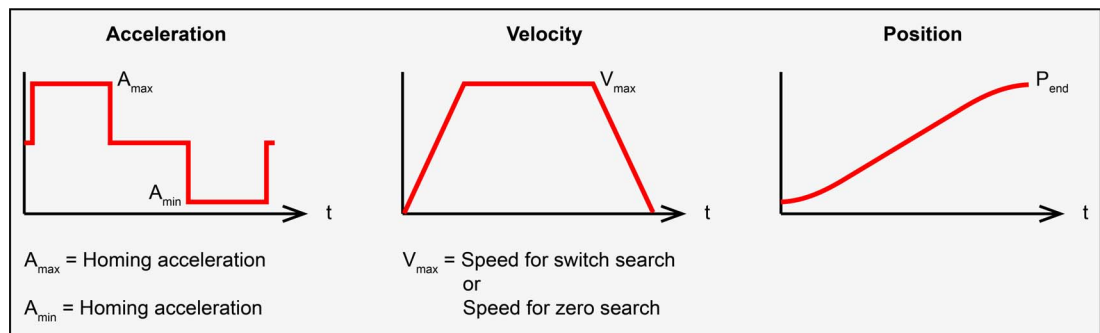


Figure 3-11 Homing trajectory – Linear ramp (trapezoidal profile)

3.5.2 How to use «HMM»

CONFIGURATION PARAMETERS

Parameter	Index	Description
→ Digital input properties	0x3141	Input polarity and logic state
→ Configuration of digital inputs	0x3142	Permits the configuration of digital inputs to digital input functionalities. Necessary for limit and homing switches that are used for «Homing».
→ Digital inputs	0x3141	Input functionality state (after polarity correction)
→ Motion profile type	0x6086	Selects the type of motion profile used for the movement: 0 = linear ramp (trapezoidal profile)

Table 3-26 Homing Mode – Configuration parameters

COMMANDING PARAMETERS

Parameter	Index	Description
→ Controlword	0x6040	The mode will be controlled by a write access to the controlword's mode-dependent bits
→ Homing method	0x6098	Defines the type of homing procedure
→ Homing speeds	0x6099	Specifies the two speeds used for Homing: In a typical cycle, the faster speed is used to find the home switch and for the offset move, the slower speed is used to find the index pulse.
→ Homing acceleration	0x609A	Specifies the acceleration during Homing
→ Home offset move distance	0x30B1	The distance to move away from a detected position upon end of the homing sequence
→ Home position	0x30B0	Allows to displace zero in the user's coordinate system
→ Current threshold for homing mode	0x30B2	The current threshold for current index homing methods

Table 3-27 Homing Mode – Commanding parameters

CONTROLWORD (HOMING MODE-SPECIFIC BITS)

Bit 15...9	Bit 8	Bit 7	Bit 6, 5	Bit 4	Bit 3...0
→ Table 6-126	Halt	→ Table 6-126	reserved	Homing operation start	→ Table 6-126

Table 3-28 Homing Mode – Controlword

Name	Value	Description
Homing operation start	0	Do not start homing procedure
	0→1	Start or continue homing procedure
Halt	0	Execute instruction of bit 4
	1	Stop axis with →Homing acceleration

Table 3-29 Homing Mode – Controlword bits

OUTPUT PARAMETERS

Parameter	Index	Description
→ Statusword	0x6041	Mode state can be observed by the statusword bits

Table 3-30 Homing Mode – Output parameters

STATUSWORD (HOMING MODE-SPECIFIC BITS)

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9...0
Position referenced to home position	➔ Table 6-127	Homing error	Homing attained	➔ Table 6-127	Target reached	➔ Table 6-127

Table 3-31 Homing Mode – Statusword

Name	Value	Description
Bit 15 Position referenced to home position	0	The position is not referenced to the home position (for example homing not yet attained or position overflow)
	1	Homing was attained and the position is referenced to the home position

Table 3-32 Homing Mode – Statusword bit 15

Bit 13 Homing error	Bit 12 Homing attained	Bit 10 Target reached	Description
0	0	0	Homing procedure is in progress
0	0	1	Homing procedure is interrupted or not started
0	1	X	Homing procedure is completed successfully
1	0	X	Homing error occurred

Table 3-33 Homing Mode – Statusword bits 10, 12, and 13

3.5.3.14 Homing Method -3 (Current Threshold Positive Speed)

The principle is similar to homing method -1 except that the →Home position is not dependent on the index pulse but only on the mechanical end stop.

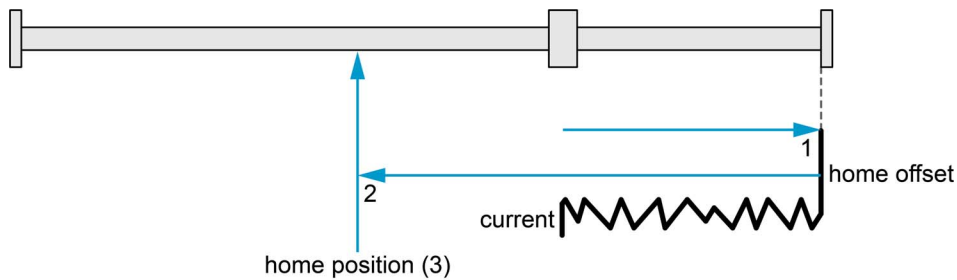


Figure 3-25 Homing method -3

3.5.3.15 Homing Method -4 (Current Threshold Negative Speed)

The principle is similar to homing method -2 except that the →Home position is not dependent on the index pulse but only on the mechanical end stop.

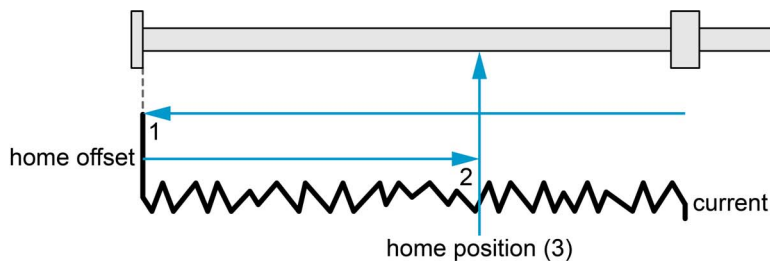


Figure 3-26 Homing method -4